

Prakrit Tyagi

Delhi, India | prakrittyagi.work@gmail.com | +91 7042037597

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY:

B.Tech in Computer Science with strong foundations in programming, data structures, algorithms, and software development. Experienced full-stack and game developer who builds scalable, user-friendly applications and innovative gameplay using modern front-/back-end stacks and major game engines; a collaborative team player focused on clean, efficient code and thoughtful design.

CORE SKILLS:

- **Languages:** C, C++, C#, Java, Python, JavaScript/TypeScript
- **Front End:** HTML, CSS, React (Hooks), Responsive/Accessibility
- **Back End:** Node.js, Express, REST/GraphQL, WebSockets
- **Databases:** MySQL, MongoDB
- **ML / Computer Vision:** Python, OpenCV (object detection, measurement, document scanning)
- **Game Development:** Unity, Unreal; gameplay systems, prototyping, optimization
- **APIs & Tools:** Twilio API, Groq (LLM), ngrok, Git/GitHub, VS Code, Eclipse
- **Operating Systems:** Windows, Linux

WORK EXPERIENCE:

Self-Employed — Indie Game Developer | India Remote | Aug 2024 – Present

- Developing an action-survival title (combat, AI, inventory/crafting, save/load)
- Building tools/pipelines; profiling & optimizing performance
- Managing roadmap, versioning, releases; shipping small side projects

Full-Stack Developer Intern — ULAVI Technologies PTE LTD | Singapore Remote | Jun 2024 – Aug 2024

- WhatsApp itinerary bot (LLM/Groq) with Twilio API
- Node.js + MySQL backend; ngrok for secure local testing
- Authored tests, debugging, and handover docs

Global Industry Immersion (Academic) — Cybersecurity Focus | USA & UK | Jan 2024 – Jun 2024

- Visits: Google, NASA; observed large-scale systems practices
- Barclays (Whippany, NJ): secure SDLC, access control, cloud basics
- Summarized learnings and applied to projects/coursework

Game Developer Intern — Khiladi Adda (Techbeliever Technologies Pvt. Ltd.) | Noida | Jan 2023 – Apr 2023

- Prototyped gameplay in Unity/C# with designers & artists
- Implemented core mechanics; improved frame time via profiling
- Documented workflows to speed iteration

Self-Employed — Backend Engineer (Game Servers) | Remote | Sep 2020 – Feb 2022

- Hosted/maintained Minecraft, Rust & other servers (Linux)
- Automated backups, patching, and migrations; Docker + Nginx deployments
- Tuned tick rate, I/O, memory; monitoring & alerts for uptime/latency

PROJECTS:

(Selected projects — more on GitHub: [GitHub](#))

Survival Action Game (Indie) — Unity (C#), State Machines (AI), Addressables, Scriptable Objects

- Core systems: real-time combat, enemy AI (patrol/chase/attack), loot, inventory/crafting, quests, save/load
- Level streaming + Addressables cut scene load times by ~30–40% on mid-range hardware
- Profiling + pooling sustained 60+ FPS @1080p in arenas with 20+ active NPCs
- Designer-friendly data via Scriptable Objects; 20+ items, 8 enemy archetypes, modular abilities
- Playtesting (n=15) improved session length by ~25% after balance pass

Snake AI Trainer — Python, RL (Double DQN), FastAPI/WebSocket, NumPy/PyTorch

- Agent outperformed rule-based baseline by ~2–3x average score after ~50 training epochs (500k+ steps)
- Prioritized replay + target network stabilized learning (beta anneal, update every 2k steps) improving sample efficiency ~1.3x
- Live dashboard charts reward/length/loss; checkpoints enable resume & A/B comparisons across seeds (n=20)
- Inference ~2–5 ms/step CPU-only with frame-stacking, reward shaping (survival + food), and early-stop guards

Video Editor (Timeline-based Desktop Tool) — FFmpeg pipeline, GPU (NVENC/NVIDIA AV1)

- Drag-drop, ripple trim, snap, markers; presets for copy-stream or GPU re-encode
- Copy-stream export: 30-min timeline → output in <60s on SSD; GPU mode cuts re-encode time ~50–70%
- Batch/CLI queue, autosave, non-destructive timeline, undo/redo, per-track gain
- Processed 100+ clips (≈ 8–10 hrs) in tests; preflight checks cut failed renders by >80%

CERTIFICATIONS:

- **CCNAv7**: Introduction to Networks — Cisco Networking Academy (Nov 2023)
- **Adobe Photoshop** — Udemy (Mar 2021)
- **Unreal Engine 4** — Udemy (Aug 2020)
- **Ethical Hacking** — Udemy (Sep 2020)

PUBLICATIONS:

- **Research Article: “Fusion of Deep Learning and Multi View Geometry for Robust Object Detection in Distributed Camera”** — *Journal of Information Systems Engineering and Management*, Vol. 10 No. 58s, e-ISSN: 2468-4376 (2025). Authors: Prakrit Tyagi, Dev Singhal, Abdul Aziz, Arushi Sajwan, Dr. Anant Kumar Jayswal — [Link to Paper](#)
- **Term Paper: “Ethical Hacking”** — Term Paper, Sem V, Amity University (2023). Authors: Prakrit Tyagi, Dr. Bhupinder Singh — [Link to Paper](#)
- **Term Paper: “Image Recognition Using Machine Learning Techniques”** — Sem III & IV, Amity University (2022). Authors: Prakrit Tyagi, Dr. Nancy Girdhar — [Link to Paper](#)

AWARDS:

- **Best Startup Potential — Technovate 2025 (Amity University)** — Project: “Fusion of Deep Learning and Multi-View Geometry for Robust Object Detection in Distributed Camera”
- **Brackeys Game Jam** — Participant — Built and submitted a Unity prototype aligned with the theme; role: gameplay/programming — Year: 2024
- **Vibe Hack 2025** — Participant — Team project delivered end-to-end; role: full-stack/game dev; pitched final demo.

EDUCATION:

- **B.Tech, Computer Science & Engineering** — Amity University, Noida — *First Division* (2025)
 - 3-Continent Program (1 semester): Birkbeck, University of London (Jan–Mar 2025); Fairleigh Dickinson University, NJ, USA (Mar–May 2025)
 - Industry Immersion (Academic): Jan–Jun 2024 — Visits to Google and NASA; Barclays (Whippany, NJ) cybersecurity briefings/workshops
 - Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Distributed Systems
- **High School Diploma** — Springdales School, Pusa Road, New Delhi

PERSONAL DETAILS:

Age: 24 Years

Residence: N-13, Second Floor, Riviera Apartment, Mall Road, Delhi – 110054, INDIA